Jadard Technology Inc.

JDTP Driver Porting Guide For MTK

2024/01/08

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Abstract

JDTP Driver Porting Guide for MTK Platform	
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Owner	Touch Team, Jadard TDDI
Classification	

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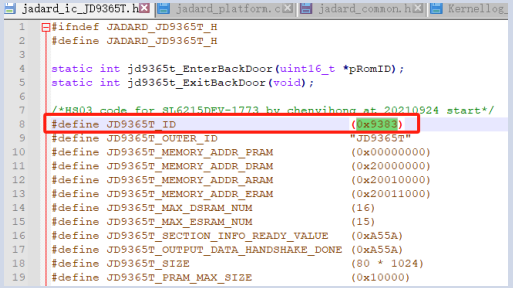
Revision History

Date	Version	List of changes	Author	Approved by
2024.01.08	1.06	Add CHIP ID Select Table	Hero.He	
2023.11.23	1.05	Add TP Suspend/Resume Process	Hero.He	
2023.10.12	1.04	Add Gesture Sequence	Hero.He	
2023.09.05	1.03	Optimize layout	Sen.Chen	
2023.08.12	1.02	Remove Proximity Component node	Hero.He	
2023.08.11	1.01	First release	Hero.He	

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JDTP CHIP ID INTRODUCTION

TDDI芯片型号	别称(曾用名)	CHIP ID	备注
JD9365T	F96368/JD9365T-B1/B2/B4/B6	0x9383	CHIP ID做为芯片的唯一识别码，是TP驱动加载的关键，因为不同是芯片，硬件寄存器，芯片设计，FW Code，均有差别。 每一份代码里设定的CHIP ID对应的都是相关的芯片型号，所以请勿去修改TP驱动文件"jadard_ic_JDxxxx.h"里的芯片ID的数值让TP驱动往下跑，否则TP会没有触摸。
JD9365TR	JD9365T-C/C1/C2/C10	0x9382	
JD9365TG	HI8561/8561N/8561P	0x9389	
JD9522T	HI8559A/8559B/8559C	0x9525	
JD9522TS-C4	HI8559TS/HI8559TB/HI8559SA-B	0x9526	
JD9366TC	无	0x9032	
JD9366TS/JD9366TP	无	0x9386	



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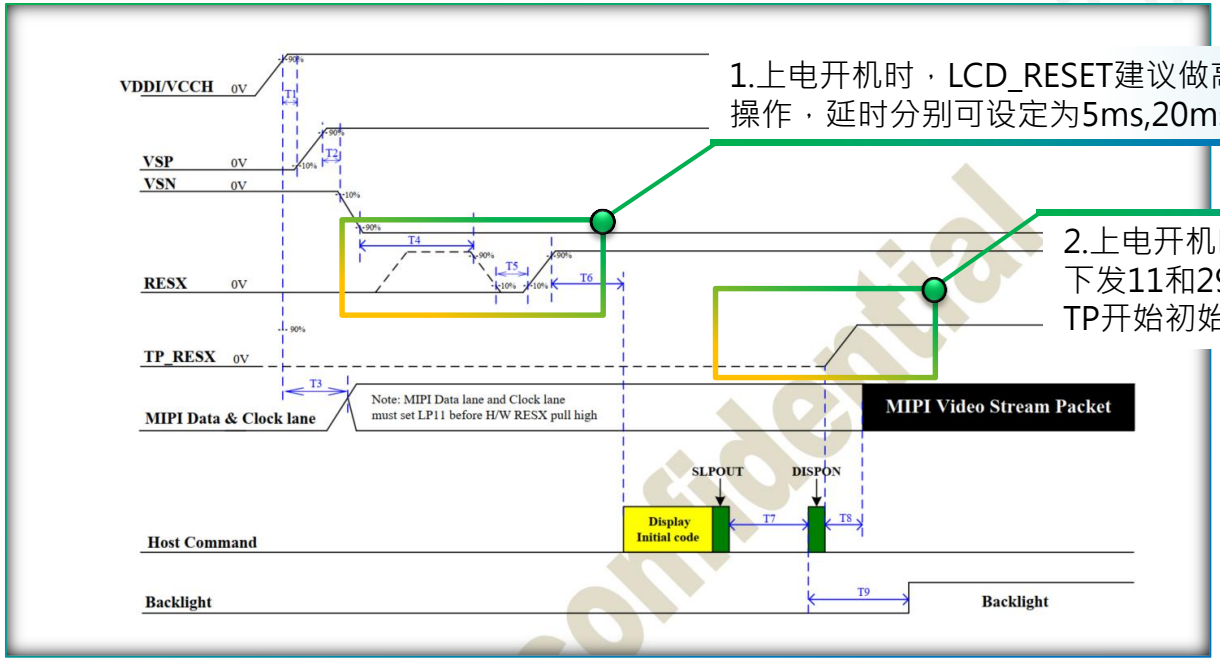
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LCD Sequence Debugging

2.1 Power On Sequence:



1.上电开机时，LCD_RESET建议做高低高操作，延时分别可设定为5ms,20ms,5ms。

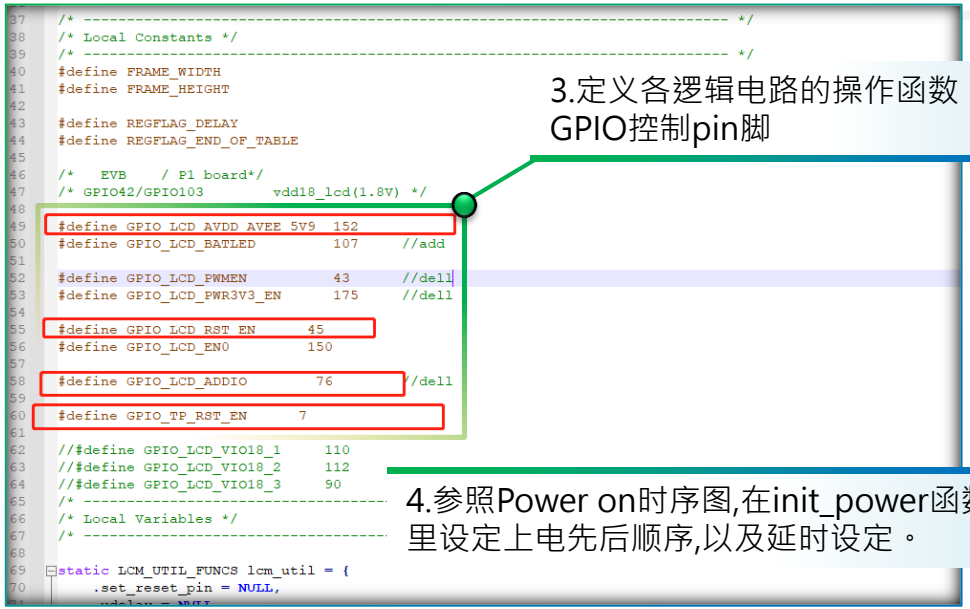
2.上电开机时，TP_RESET应延时拉高，等下发11和29后，再把TP_RESET拉HIGH，TP开始初始化。

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LCD Sequence Debugging

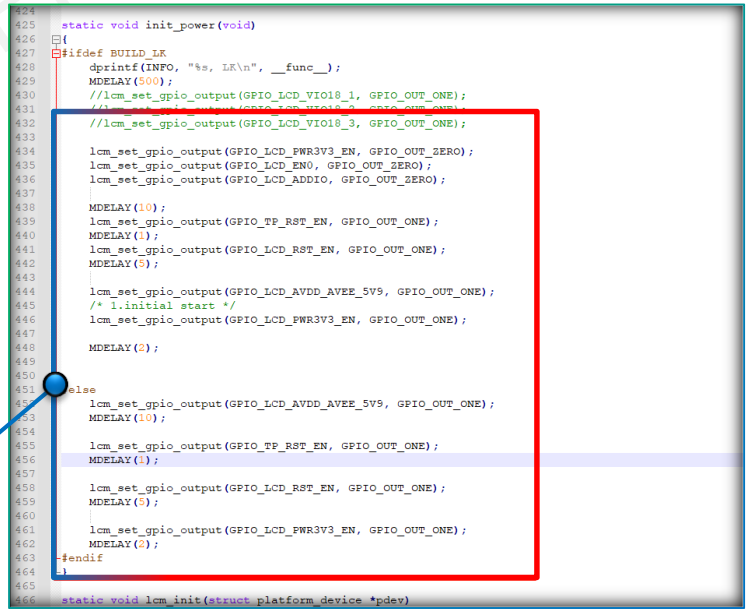
LCD Power On Sequence Debug Method:

在...\vendor\mediatek\proprietary\bootable\bootloader\lk\dev\lcm的目录下找到屏的*.c文件，在init_power函数里配置IOVCC(VDDI)/AVDD&AVEE(VSP&VSN)/LCD_RESET/TP_RESET上电时序。如下图：



3.定义各逻辑电路的操作函数，GPIO控制pin脚

4.参照Power on时序图,在init_power函数里设定上电先后顺序,以及延时设定。

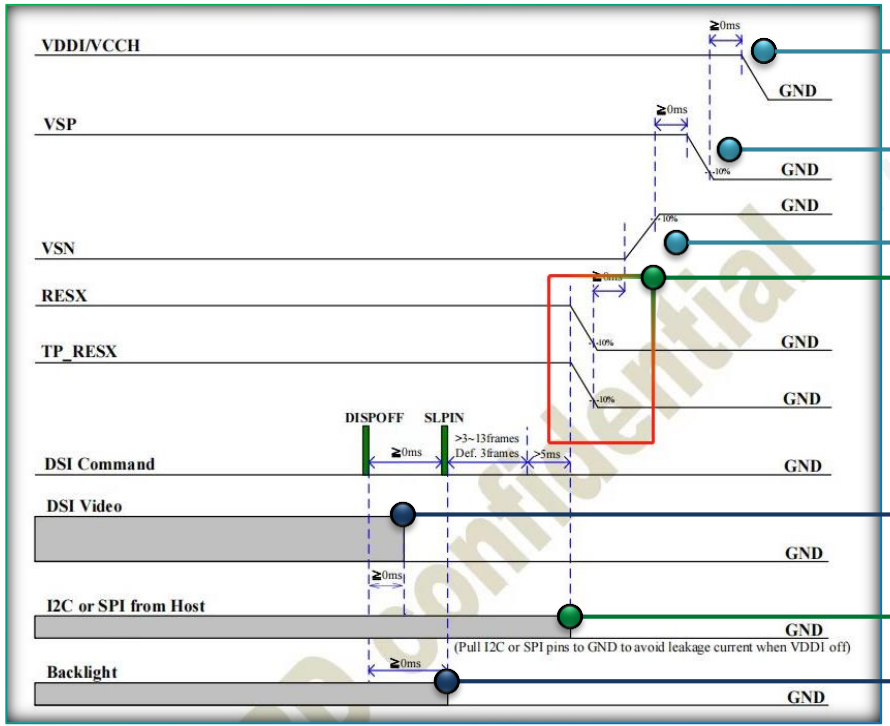


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LCD Sequence Debugging

2.2 Power Off Sequence:



Step 3 再接着是AVDD/AVEE下电,最后VDDI下电。

Step 2 LCD_RESET和TP_RESET、IIC&SPI下电。

Step 1 MIPI最先下电，然后关背光(Backlight)。

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LCD Sequence Debugging

LCD Power Off Sequence Debug Method:

1. 在...\kernel-4.14\driver\misc\mediatek\lcm\lcm_pmic.c文件里按照规格书的时序图配置即可



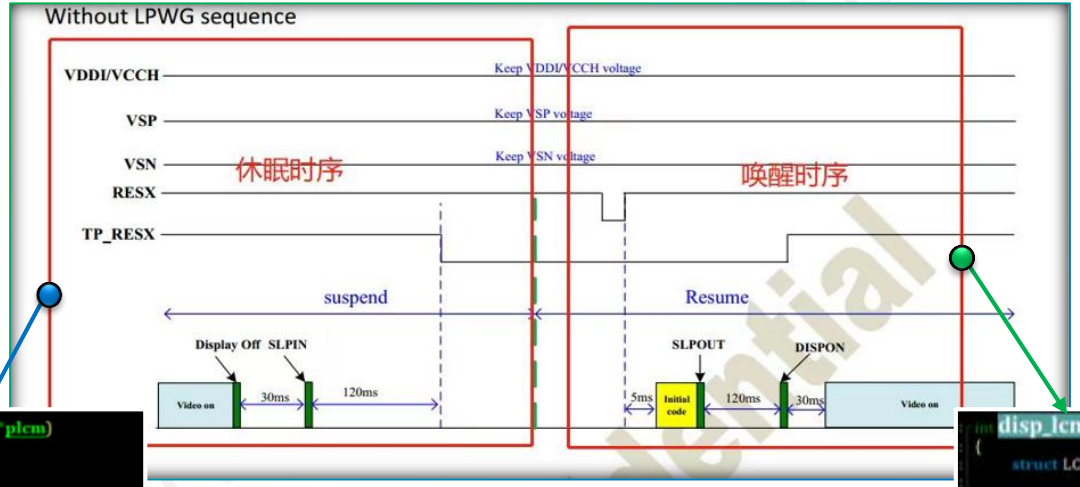
2. 关机下电时序调试步骤：
   1、首先是mipi的动作，先跑Display off(28)指令和Sleep in(10)指令，然后进入LP模式最后下拉.
   2、I2C/SPI在mipi进入Sleep in模式后也会由TP驱动去给TP FW下发进入睡眠模式(TP Suspend).
   3、LCD reset和TP reset都相继下拉，使IC退出工作模式.
   4、VSN(AVEE)下拉.
   5、VSP(AVDD)下拉，3 power下电完成。如果是4 power的话，会有个VDDD电压，然后VDDD下拉.
   6、最后IOVCC下拉，IC停止工作.

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LCD Sequence Debugging

2.3 LCD Suspend/Resume Sequence:




```
disp_lcm_suspend(struct disp_lcm_handle *plcm)
{
    struct LCM_DRIVER *lcm_drv = NULL;

    DISPFUNC();
    if (_is_lcm_inited(plcm)) {
        lcm_drv = plcm->drv;
        if (lcm_drv->suspend) {
            lcm_drv->suspend();
        } else {
            DISPERR("FATAL ERROR, lcm_drv->suspend is null\n");
            return -1;
        }

        if (lcm_drv->suspend_power)
            lcm_drv->suspend_power();


        return 0;
    }
    DISPERR("lcm_drv is null\n");
```

```
disp_lcm_resume(struct disp_lcm_handle *plcm)
{
    struct LCM_DRIVER *lcm_drv = NULL;

    DISPFUNC();
    if (_is_lcm_inited(plcm)) {
        lcm_drv = plcm->drv;

        if (lcm_drv->resume_power)
            lcm_drv->resume_power();


        if (lcm_drv->resume) {
            lcm_drv->resume();
        } else {
            DISPERR("FATAL ERROR, lcm_drv->resume is null\n");
            return -1;
        }

        return 0;
    }
    DISPERR("lcm_drv is null\n");
```

LCD Sequence Debugging

LCD Suspend/Resume Debug Method:

1. 在...\vendor\mediatek\proprietary\bootable\bootloader\lk\dev\lcm目录内,屏的*.c文件里找到相关的函数

```
static void lcm_suspend(void)
{
unsigned int data_array[16];
    data_array[0] = 0x00280500;
    dsi_set_cmdq(data_array, 1, 1);
    MDELAY(20);

    data_array[0]=0x00100500;
    dsi_set_cmdq(data_array, 1, 1);
    MDELAY(120);

    lcm_set_gpio_output(GPIO_TPD_RST, 0);
    MDELAY(10);
```

1. lcm_suspend函数里添加TP_RESET的拉LOW操作

```
static void lcm_init_lcm(void)
{
    //lcm_set_gpio_output(GPIO_LCD_PWR, 1);
        lcm_set_gpio_output(GPIO_LCD_AF, 1);
        MDELAY(5);
        //lcm_set_gpio_output(GPIO_LCD_PWR, 1);
        lcm_set_gpio_output(GPIO_LCD_18_PWR, 1);
        MDELAY(10);
        lcm_set_gpio_output(GPIO_LCD_33_PWR, 1);
        MDELAY(5);
        display_bias_enable();
        MDELAY(30);
        lcm_set_gpio_output(GPIO_LCD_RST, 1);
        MDELAY(10);
        lcm_set_gpio_output(GPIO_LCD_RST, 0);
        MDELAY(20);
        lcm_set_gpio_output(GPIO_LCD_RST, 1);
        MDELAY(120);

    push_table(lcm_initialization_setting, sizeof(lcm_initializatio
        MDELAY(20);
        lcm_set_gpio_output(GPIO_TPD_RST, 1);
```

2. lcm_init函数(resume跑初始化)里添加TP_RESET的拉HIGH操作

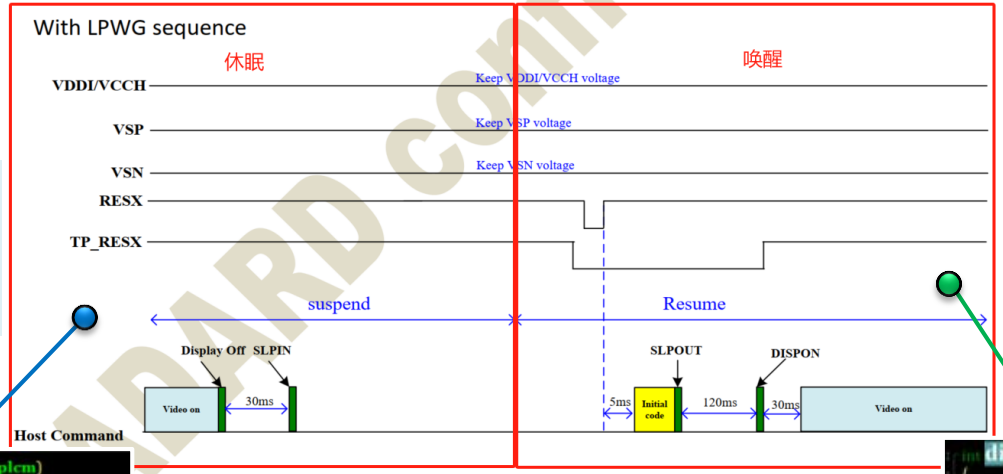
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LCD Sequence Debugging

2.4 LCD Suspend/Resume Sequence(With Gesture):

带手势唤醒功能时,
lcm_suspend函数里需保持
AVDD/AVEE/LCD-RST/
/TP-RST/均不下电




```
int disp_lcm_suspend(struct disp_lcm_handle *plcm)
{
    struct LCM_DRIVER *lcm_drv = NULL;

    DISPFUNC();
    if (_is_lcm_inited(plcm)) {
        lcm_drv = plcm->drv;
        if (lcm_drv->suspend) {
            lcm_drv->suspend();
        } else {
            DISPERR("FATAL ERROR, lcm_drv->suspend is null\n");
            return -1;
        }

        if (lcm_drv->suspend_power)
            lcm_drv->suspend_power();

        return 0;
    }
    DISPERR("lcm_drv is null\n");
```

```
int disp_lcm_resume(struct disp_lcm_handle *plcm)
{
    struct LCM_DRIVER *lcm_drv = NULL;

    DISPFUNC();
    if (_is_lcm_inited(plcm)) {
        lcm_drv = plcm->drv;

        if (lcm_drv->resume_power)
            lcm_drv->resume_power();

        if (lcm_drv->resume) {
            lcm_drv->resume();
        } else {
            DISPERR("FATAL ERROR, lcm_drv->resume is null\n");
            return -1;
        }

        return 0;
    }
    DISPERR("lcm_drv is null\n");
```

LCD Sequence Debugging

LCD Suspend/Resume(With Gesture) Debug Method:

1. 在...\vendor\mediatek\proprietary\bootable\bootloader\lk\dev\lcm目录内,屏的*.c文件里找到相关的函数

```
static void lcm_suspend(void)
{
    unsigned int data_array[16];

#ifdef BUILD_LK
    dprintf(INFO, "%s, LK\n", __func__);
#else
    pr_debug("%s, kernel", __func__);
#endif

#ifdef PUSH_TABLET_USING
    push_table(lcm_suspend_setting, sizeof(lcm_suspend_setting) / sizeof(struct LCM_setting_table), 1);
#else
    data_array[0] = 0x00280500;
    dsi_set_cmdq(data_array, 1, 1);
    MDELAY(10);

    data_array[0] = 0x00100500;
    dsi_set_cmdq(data_array, 1, 1);
    MDELAY(120);
#endif

    lcm_set_gpio_output(GPIO_TP_RST_EN, GPIO_OUT_ONE);//TP-RESET拉高

    //lcm_set_gpio_output(GPIO_LCD_PWR3V3_EN, GPIO_OUT_ZERO);
    MDELAY(10);
    //lcm_set_gpio_output(GPIO_LCD_VIO18_1, GPIO_OUT_ZERO);
    //lcm_set_gpio_output(GPIO_LCD_VIO18_2, GPIO_OUT_ZERO);
    //lcm_set_gpio_output(GPIO_LCD_VIO18_3, GPIO_OUT_ZERO);
}
```

1. lcm_suspend函数里保持TP_RESET为High

```
static void lcm_init(struct platform_device *pdev)
{
#ifdef BUILD_LK
    dprintf(INFO, "%s, LK\n", __func__);
#else
    lcm_request_gpio_control();
#endif
    lcm_set_gpio_output(GPIO_LCD_AVDD_AVEE_5V9, GPIO_OUT_ONE);
    MDELAY(25);

    lcm_set_gpio_output(GPIO_LCD_RST_EN, GPIO_OUT_ONE);
    MDELAY(25);
    lcm_set_gpio_output(GPIO_LCD_RST_EN, GPIO_OUT_ZERO);
    MDELAY(25);
    lcm_set_gpio_output(GPIO_LCD_RST_EN, GPIO_OUT_ONE);
    MDELAY(25);
    lcm_set_gpio_output(GPIO_LCD_RST_EN, GPIO_OUT_ZERO);
    MDELAY(1);
    //lcm_set_gpio_output(GPIO_TP_RST_EN, GPIO_OUT_ZERO);//TP-RESET拉低
    MDELAY(25);
    lcm_set_gpio_output(GPIO_LCD_RST_EN, GPIO_OUT_ONE);
    MDELAY(25);

    lcm_set_gpio_output(GPIO_LCD_PWR3V3_EN, GPIO_OUT_ONE);
    lcm_set_gpio_output(GPIO_LCD_EN0, GPIO_OUT_ONE);
    lcm_set_gpio_output(GPIO_LCD_ADDIO, GPIO_OUT_ONE);

    MDELAY(120);
    lcm_init_register();
    MDELAY(20);
    lcm_set_gpio_output(GPIO_TP_RST_EN, GPIO_OUT_ZERO);
    MDELAY(5);
    lcm_set_gpio_output(GPIO_TP_RST_EN, GPIO_OUT_ONE);//DISPLAY下完Code后,TP-RESET拉高低高
    MDELAY(1);
```

2. lcm_init函数(resume跑初始化)里添加TP_RESET的拉HIGH操作

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TP Suspend and TP Resume Process

2.5 TP Suspend and TP Resume Process:

1.先将TP Suspend/Resume function 和系統的休眠唤醒串接起来，注意系统是使用何种方式进行休眠唤醒的。

```
static const struct i2c_device_id jadard_common_ts_id[] = {
    {JADARD_common_NAME, 0 },
    {}
};

static const struct dev_pm_ops jadard_common_pm_ops = {
#if (!defined(JD_CONFIG_FB)) && (!defined(JD_CONFIG_DRM))
    .suspend = jadard_common_suspend,
    .resume  = jadard_common_resume,
#endif
};

#ifdef CONFIG_OF
static struct of_device_id jadard_match_table[] = {
    {.compatible = "jadard,jdcommon" },
    {},
};
#else
#define jadard_match_table NULL
#endif

static struct i2c_driver jadard_common_driver = {
    .id_table    = jadard_common_ts_id,
    .probe       = jadard_chip_common_probe,
    .remove      = jadard_chip_common_remove,
    .driver      = {
        .name = JADARD_common_NAME,
        .owner = THIS_MODULE,
        .of_match_table = jadard_match_table,
#ifdef CONFIG_PM
        .pm             = &jadard_common_pm_ops,
#endif
    },
};

#if defined(__JADARD_KMODULE__)
int jadard_common_init(void)
```

jadard_common.h

```
146  *  Customer/platform config
147  */
148 /* #define JD_OPPO_FUNC */
149
150 /*
151  *  Suspend/Resume function
152  *
153  * Frame Buffer (FB): JD_CONFIG_FB
154  * Direct Rendering Manager (DRM): JD_CONFIG_FB + JD_CONFIG_DRM
155  * Qualcomm MSM DRM: JD_CONFIG_FB + JD_CONFIG_DRM_MSM
156  * MTK Platform DRM: JD_CONFIG_SPRD_DRM
157  * System Node: JD_CONFIG_NODE
158  */
159 /* #define JD_CONFIG_FB */
160 /* #define JD_CONFIG_DRM */
161 /* #define JD_CONFIG_DRM_MSM */
162 /* #define JD_CONFIG_SPRD_DRM */
163 /* #define JD_CONFIG_NODE */
164
165 /*
166  *  Enable I2C single mode for with flash only
167  */
168 /* #define JD_I2C_SINGLE_MODE */
169
170 /*
171  *  Enable each workarounds for FPGA only
172  */
173 /* #define JD_FPGA_WORKAROUND */
174 /*========== Config select end ==========*/
175
```

2. 在TP驱动的jadard_common.h文件中开启相关的宏定义。

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TP Suspend and TP Resume Process

2.5.1 TP Suspend and TP Resume Process:

TP Suspend放到屏下电前，关闭TP-irq，
TP Resume时先上电再开启TP-irq。

```
105.478746] [MIPI]>>>mipi_common_panel_disable(638):disable, delay_ms = 20ms.
105.478793] [MIPI]>>>-mipi_common_panel_disable(642).
105.478827] [MIPI]>>>+mipi_common_panel_unprepare(652).
105.478891] [MIPI]>>>+mipi_common_panel_off(595).
105.478958] [MIPI]>>>-mipi_common_panel_off(613).
105.495423] [MIPI]>>>+mipi_common_reset_set(408):gpio rst-gpio(103), flag(0), panelId(0x12).
105.495485] [MIPI]>>>mipi_common_reset_set(466):after sleep 0ms, gpio rst-gpio(103) output 0.
105.495511] [MIPI]>>>-mipi_common_reset_set(480).
105.525607] [MIPI]>>>mipi_common_panel_unprepare(681):Disable s5v gpio.
105.542274] [MIPI]>>>mipi_common_panel_unprepare(690):Disable avdd gpio.
105.558941] [MIPI]>>>mipi_common_panel_unprepare(697):Panel supply is null, to get it.
105.559547] mipi-common ff450000.dsi.0: Linked as a consumer to regulator.17
105.559600] [MIPI]>>>mipi_common_panel_unprepare(709):Disable power supply.
105.575877] [MIPI]>>>mipi_common_panel_unprepare(744):Power supply disable successfully.
105.702276] [MIPI]>>>mipi_common_panel_unprepare(753):unprepare, delay_ms = 120ms.
105.702351] [MIPI]>>>-mipi_common_panel_unprepare(757).
105.702421] <DW MIPI DSI-INFO>dw_mipi_dsi_encoder_disable(1159).
105.708058] techpoint 2-0045: techpoint_suspend
105.788959] techpoint 2-0045: detect_thread_function: exit
105.789161] techpoint 2-0045: detect_thread_stop ...
105.789571] imotor mmc_pwrseq_simple_suspend-292
105.789754] <BD37033-INFO>bd37033_i2c_suspend(3153).
105.789839] fm get power, power status:0
105.789939] [JDTP] jadard_chip_common_suspend: enter
105.789984] [JDTP][IMOTOR]set gpio12, value=0.!!!
105.790009] [JDTP] jadard_chip_common_suspend: reset pin = 12
105.790050] [JDTP] irq_enable = 0
105.790083] [JDTP] jadard_chip_common_suspend: end
105.791035] rk808-rtc rk808-rtc: RTC date/time 2023-11-16(4) 13:03:26
105.799435] [imotor-gpio][INFO][imotor_gpios_suspend:109]  Enter (sleep gpio).
105.799526] [WLAN_RFKILL]: Enter rfkill_wlan_suspend
```

屏全下电方案

TP休眠

3.在某些平台上整机休眠是采用全断电的方式时，TP的TP Suspend会跑到下电之后，导致中断没有处理完，一直拉住IIC。这时需把TP的suspend提前至断电之前进行。或者是在断电前休眠前拉低TP-irq。

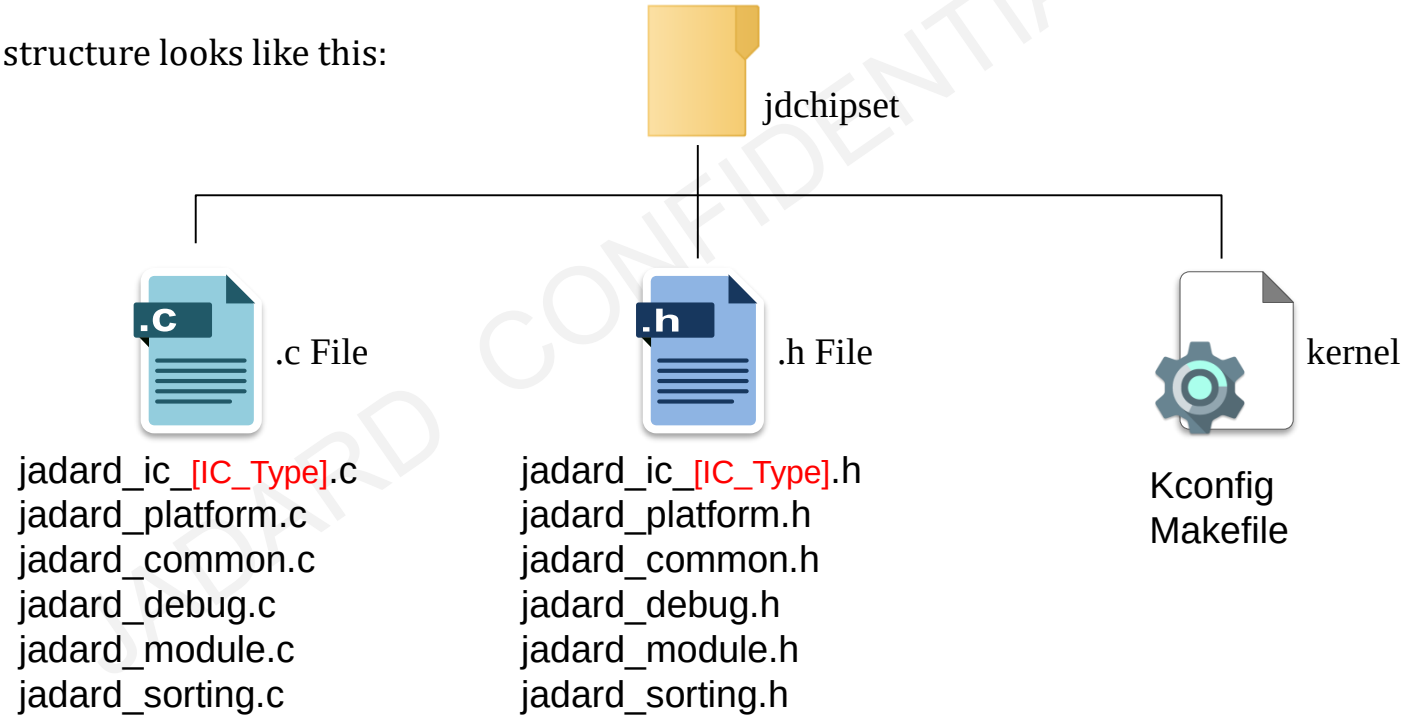
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Driver Code File Structure Introduction

The directory of "Jadard TP Driver" is named as "jdchipset" as default, that means all driver files are in it.

The directory structure looks like this:



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Driver Code File Structure Introduction

The Detail Description Of Touch Driver Files:

Component	File	Attribute	Description
Main	jadard_ic_[IC_Type].c jadard_ic_[IC_Type].h jadard_common.c jadard_common.h	Required	The main function of TP driver, that containing driver registration, report point, suspend/resume, gesture function,multi-touch protocol support and so on
Module Init	jadard_platform.c jadard_platform.h	Required	Bus interface (SPI/I2C) initialization
ADB Debug	jadard_debug.c jadard_debug.h jadard_module.c jadard_module.h	Required	Create /proc/jadard_touch node, use to communicate with APK or ADB.
Factory Test	jadard_sorting.c jadard_sorting.h	Required	Use for factory test; recommend to use APK for factory test
Complier	Makefile Kconfig	Required	Use for kernel compiling and configuration

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Porting TP Driver to MTK Platform

4.1 Copy Driver Files Into Kernel:

At first, copy "jdchipset" directory into kernel directory (…/kernel/drivers/input/touchscreen/).
Then modify Kconfig/Makefile:

a. Add the line below to …/kernel/drivers/input/touchscreen/**Kconfig**:
   source "drivers/input/touchscreen/jdchipset/Kconfig"

b. Add the line below to …/kernel/drivers/input/touchscreen/**Makefile**:
   obj-$(CONFIG_TOUCHSCREEN_JADARD_CHIPSET) += jdchipset/

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Porting TP Driver to MTK Platform

4.2 Interface Setting Recommendation:

I2C interface

- I2C speed recommendation: 400K
- I2C address : 0x68 (7 bit)

SPI interface

Different platforms have different settings, please refers to the following recommendations:

IC Serials	SPI Mode	SPI Speed	Description
JD9365T/JD9522T/JD9366TC	Mode 0 / Mode 1	6M ~ 10M	MTK/全志/杰发

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Kernel Configuration

4.3 Kernel Config Add The Following:

@ kernel/arch/arm64/config/xxxx_deconfig

---Add:

CONFIG_TOUCHSCREEN_JADARD_CHIPSET=y

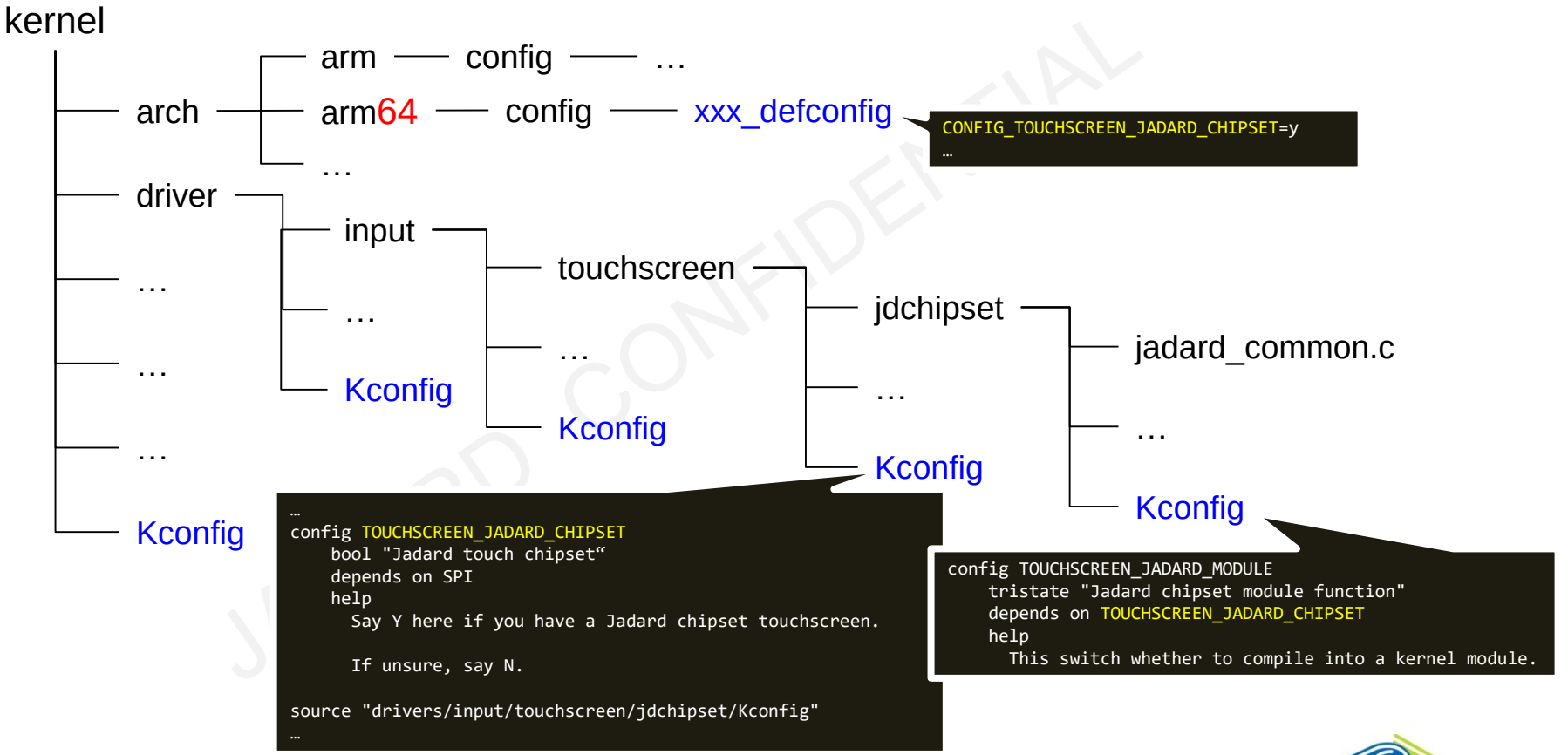
@ /kernel/drivers/input/touchscreen/jdchipset/Makefile

---Add :

```
obj-y       += jadard_module.o
obj-y       += jadard_ic_JD9366T.o
obj-y       += jadard_common.o  jadard_platform.o
obj-y       += jadard_debug.o
obj-y       += jadard_sorting.o
ccflags-y += -DCONFIG_TOUCHSCREEN_JADARD_DEBUG=y
ccflags-y += -DCONFIG_TOUCHSCREEN_JADARD_SORTING=y
```

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Kernel Configuration



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DTS Configuration

4.4 Configure DTS:

MTK DTS File Path:

…/kernel/arch/arm64/boot/dts/mediatek/k6877v1_64_pre.dts ---(MTK6877 DEMO Board)

I2C Interface Example:

As Following: Device Tree Configuration of I2C

SPI Interface Example:

As Following: Device Tree Configuration of SPI

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Device Tree Configuration of I2C

```
jadard@68 {
    compatible = "jadard,jdcommon";
    reg = <0x68>;
    status = "okay";
    interrupt-parent = <&tlmm>;
    interrupts = <125 0x2800>;
    jadard,panel-sense-nums = <30 48>;
    jadard,panel-coords = <0 1200 0 2000>;
    jadard,panel-max-points = <10>;
    jadard,int-is-edge = <1>;
    jadard,irq-gpio = <&tlmm 125 0x2008>;
    jadard,rst-gpio = <&tlmm 89 0x00>;
};
```

→ Sense number及Resolution需依照實際Panel數值調整

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Device Tree Configuration of SPI

```
jadard_ts@0 {
    compatible = "jadard,jdcommon";
    reg = <0x0>;
    status = "OK" ;
    spi-max-frequency = <6000000>;
    interrupt-parent = <&ap_gpio>;
    interrupts = <144 IRQ_TYPE_EDGE_FALLING 144 0>;
    jadard,panel-sense-nums = <30 48>;
    jadard,panel-coords = <0 1200 0 2000>;
    jadard,panel-max-points = <10>;
    jadard,int-is-edge = <1>;
    jadard,irq-gpio = <&ap_gpio 144 GPIO_ACTIVE_HIGH>;
    jadard,rst-gpio = <&ap_gpio 145 GPIO_ACTIVE_HIGH>;
};
```

→ Sense number及Resolution需依照實際Panel數值調整

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Compile kernel and generate bootimage

4.5 Compiled Code:

- Step1: source build/envsetup.sh
- Step2: lunch full_k6877v1_64_pre-userdebug
- Step3: make bootimage -j4

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JDTP System Nodes

5.1 JDTP Nodes Path:

When the device is powered on, TP driver is loaded.

Then the JADARD's nodes will be generated under "/proc/jadard_touch/" directory, For example:

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
-rw-r--r-- 1 root root 0 1970-01-01 03:15 display
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_dump
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_package
-rw-r--r-- 1 root root 0 1970-01-01 03:15 int_en
-rw-r--r-- 1 root root 0 1970-01-01 03:15 register
-rw-r--r-- 1 root root 0 1970-01-01 03:15 report_debug
--w------- 1 root root 0 1970-01-01 03:15 reset
-r--r--r-- 1 root root 0 1970-01-01 03:15 sorting_test
```

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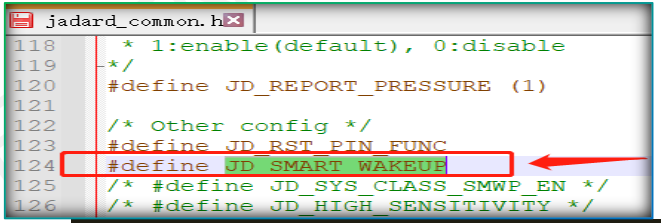
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JDTP System Nodes

5.2 Component Nodes:

Gesture Component:

"GESTURE / SMWP" which are generated when define "JD_SMART_WAKEUP" into "jadard_common.h" file.



"SMWP" Usage:

```
> adb shell
# cd proc/jadard_touch
# echo 0 > SMWP      /* Disable gesture mode*/
# echo 1 > SMWP      /* Enable gesture mode*/
# cat SWMP           /* Show current gesture status*/
```

"GESTURE" Usage:

```
> adb shell
# cd proc/jadard_touch
# echo 0011 > GESTURE   /* Enable gesture 2 & 3 */
                        /* Disable gesture 0 & 1 */
# cat GESTURE           /* Get each gesture switch status*/
```

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
-rw-r--r-- 1 root root 0 1970-01-01 03:15 display
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_dump
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_package
-rw-r--r-- 1 root root 0 1970-01-01 03:15 int_en
-rw-r--r-- 1 root root 0 1970-01-01 03:15 register
-rw-r--r-- 1 root root 0 1970-01-01 03:15 report_debug
--w------- 1 root root 0 1970-01-01 03:15 reset
-r--r--r-- 1 root root 0 1970-01-01 03:15 sorting_test
```

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JDTP System Nodes

5.3 Debugging Nodes:

JDTP's debugging nodes are used to get TP driver information, and debug TP driver, which makes us debug TP driver easier and more convenient. As following introduction to how to use JDTP's debugging nodes.

"debug" Usage:

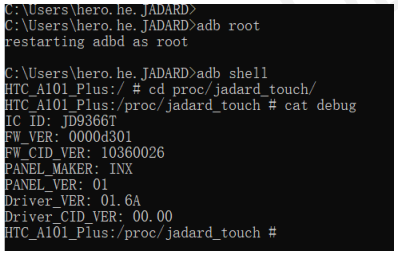
> adb shell

cd proc/jadard_touch

echo v > debug /* default */

cat debug

/*Show TP firmware version information*/





"debug" Usage:

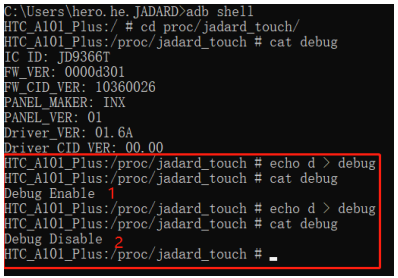
> adb shell

cd proc/jadard_touch

echo d > debug

cat debug

/* Enable/Disable JDTP debug log */



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JDTP System Nodes

"int_en" Usage:

> adb shell
cd proc/jadard_touch
echo 0 > int_en /* Disable TP interrupt*/
echo 1 > int_en /* Enable TP interrupt*/
cat int_en /* Show TP interrupt status*/

"reset" Usage:

> adb shell
cd proc/jadard_touch
echo 2 > reset /* TP reset control*/

※ "reset" node only support write

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
-rw-r--r-- 1 root root 0 1970-01-01 03:15 display
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_dump
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_package
-rw-r--r-- 1 root root 0 1970-01-01 03:15 int_en
-rw-r--r-- 1 root root 0 1970-01-01 03:15 register
-rw-r--r-- 1 root root 0 1970-01-01 03:15 report_debug
--w------- 1 root root 0 1970-01-01 03:15 reset
-r--r--r-- 1 root root 0 1970-01-01 03:15 sorting_test
```

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JDTP System Nodes

"register" Usage:

> adb shell

cd proc/jadard_touch

echo w:x40008004:xa5 > register /* Write TP register command */

echo r:x4000800c:x04 > register /* Read TP register command */

cat register /* Show the register's values */

CPU Soft Reset	echo w:x40008012:xa5 > register
System Soft Reset	echo w:x40008004:xa5 > register
讀取CPU計數器	echo r:x4000800c:x04 > register
確認AC Mode (USB)	echo r:x20002ffc:x01 > register
確認耳機插拔	echo r:x20003016:x01 > register

※紅字處以JD9365T為範例，實際位址依參考文件為準

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
-rw-r--r-- 1 root root 0 1970-01-01 03:15 display
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_dump
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_package
-rw-r--r-- 1 root root 0 1970-01-01 03:15 int_en
-rw-r--r-- 1 root root 0 1970-01-01 03:15 register
-rw-r--r-- 1 root root 0 1970-01-01 03:15 report_debug
--w------- 1 root root 0 1970-01-01 03:15 reset
-r--r--r-- 1 root root 0 1970-01-01 03:15 sorting_test
```

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JDTP System Nodes

"sorting_test" Usage:

We can use this node to run factory test and get the test results.

1. Push test configuration file(xxx_test.ini) into /vendor/firmware/
   > adb root
   > adb remount
   > adb push xxx_test.ini /vendor/firmware/
2. Use adb command to run factory test
   > adb shell
   # cd proc/jadard_touch
   # cat sorting_test /* run factory test */
3. Get test results
   Wait test finished, we can show test results PASS or FAIL.
   Then test results will be generated under /data/ directory. Use the following command to get test data.
   > adb pull /data/xxx_xxxx_test_data.csv d:\
   > adb pull /data/xxx_xxxx_test_data.txt d:\

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
-rw-r--r-- 1 root root 0 1970-01-01 03:15 display
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_dump
-rw-r--r-- 1 root root 0 1970-01-01 03:15 fw_package
-rw-r--r-- 1 root root 0 1970-01-01 03:15 int_en
-rw-r--r-- 1 root root 0 1970-01-01 03:15 register
-rw-r--r-- 1 root root 0 1970-01-01 03:15 report_debug
--w------- 1 root root 0 1970-01-01 03:15 reset
-r--r--r-- 1 root root 0 1970-01-01 03:15 sorting_test
```

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JDTP System Nodes

Upgrade Firmware Manually In The Machine:

1. 升级过程中设备请保持亮屏状态
2. adb root
3. adb remount /* 整机需要有remount权限 */
4. adb push /xxxx /JD9XXXT_xxxxx_xxx.bin /vendor/firmware/TP001.bin /*TP固件放在指定目录*/
5. adb shell "echo t TP001.bin > /proc/jadard_touch/debug" /*TP固件升级指令*/
6. adb shell "cat /proc/jadard_touch/debug" /*查看TP固件是否升级成功*/
   - 如显示update complete，请休眠唤醒一下整机，再进行测试。
   - 如显示update failed，请重启整设备。

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
…
```

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JDTP System Nodes

Check FW Version:

1. 设备请保持亮屏状态
2. adb root
3. adb shell "echo v > /proc/jadard_touch/debug" /*检查TP固件版本号指令*/
4. adb shell "cat /proc/jadard_touch/debug" /*获得TP固件版本号*/

Check Touch IC Information:

1. 设备请保持亮屏状态
2. adb root
3. adb shell "echo i > /proc/jadard_touch/debug" /*检查IC信息指令*/
4. adb shell "cat /proc/jadard_touch/debug" /*获得分辨率信息,TP通道数等*/

```
C:\Users\hero.he.JADARD>adb root
adbd is already running as root

C:\Users\hero.he.JADARD>adb shell "echo v > /proc/jadard_touch/debug"

C:\Users\hero.he.JADARD>adb shell "cat /proc/jadard_touch/debug"
IC ID: JD9366T
FW_VER: 0000d301
FW_CID_VER: 10360026
PANEL_MAKER: INX
PANEL_VER: 01
Driver_VER: 01.6A
Driver_CID_VER: 00.00

C:\Users\hero.he.JADARD>adb shell "echo i > /proc/jadard_touch/debug"

C:\Users\hero.he.JADARD>adb shell "cat /proc/jadard_touch/debug"
Jadard Touch IC Information: JD9366T
Driver register Interrupt: EDGE TIRGGER
Protocol: TYPE_B
Freq_Band: 0
X_Num: 30
Y_Num: 48
X_Resolution: 1200
Y_Resolution: 2000
Max Points: 10

C:\Users\hero.he.JADARD>
```

```
root@msm8996:/ # ls -l /proc/jadard_touch
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 GESTURE
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SENSITIVITY
-rw-rw-rw- 1 root root 0 1970-01-01 03:15 SMWP
-r--r--r-- 1 root root 0 1970-01-01 03:15 attn
-rw-r--r-- 1 root root 0 1970-01-01 03:15 debug
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag
-rw-r--r-- 1 root root 0 1970-01-01 03:15 diag_arr
…
```

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Appendices: Compile Failed

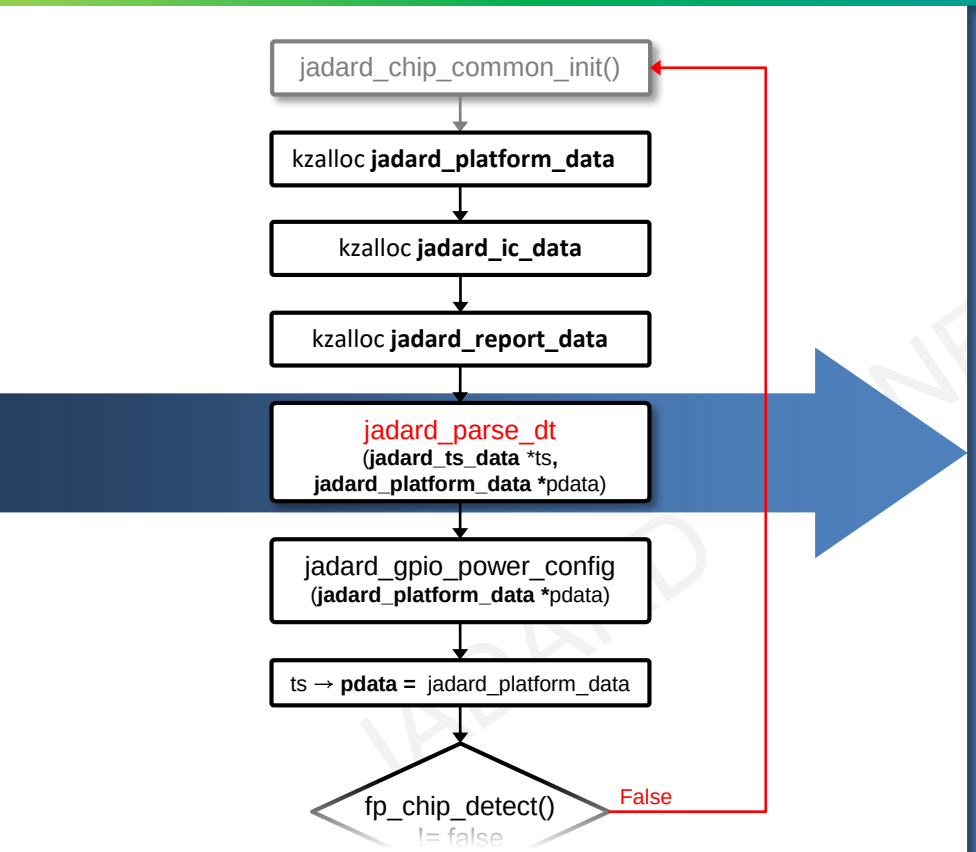
- **检查 /kernel/drivers/input/touchscreen/jdchipset/Makefile**
  - 加上下图红色框处设定

```
# Makefile for the Jadard touchscreen drivers.
obj-y       += jadard_module.o
obj-y       += jadard_ic_JD9366T.o
obj-y       += jadard_common.o jadard_platform.o
obj-y       += jadard_debug.o
obj-y       += jadard_sorting.o

ccflags-y += -DCONFIG_TOUCHSCREEN_JADARD_DEBUG=y
ccflags-y += -DCONFIG_TOUCHSCREEN_JADARD_SORTING=y
```

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Appendices: DTS Confirm



JDTP MTK Platform.dtsi

```
i2c@75ba000 {
    jadard@68 {
            compatible = "jadard,jdcommon";
            reg = <0x68>;
            interrupt-parent = <&tlmm>;
            interrupts = <125 0x2800>;
            jadard,panel-sense-nums = <32 48>;
            jadard,panel-coords = <0 1200 0 2000>;
            jadard,panel-max-points = <10>;
            jadard,int-is-edge = <1>;
            jadard,irq-gpio = <&tlmm 125 0x2008>;
            jadard,rst-gpio = <&tlmm 89 0x00>;
    };
};
```

Linux Kernel log

```
[    0.534650]  (5)[1:swapper/0][JDTP] jadard_jd9366t_init
[    2.354180]  (0)[101:kworker/u16:1][JDTP] jadard_chip_common_init: enter
[    2.354193]  (0)[101:kworker/u16:1][JDTP] DT:MAX_PT = 10, INT_IS_EDGE = 1
[    2.354204]  (0)[101:kworker/u16:1][JDTP] DT:panel-sense-num = 32, 48
[    2.354216]  (0)[101:kworker/u16:1][JDTP] DT:panel-coords = 0, 1200, 0, 2000
[    2.354227]  (0)[101:kworker/u16:1][JDTP] DT:gpio_irq = 1, gpio_rst = 0
[    2.460587]  (0)[101:kworker/u16:1][JDTP] mtk_tpd: jadard reset over
[    2.461914]  (1)[101:kworker/u16:1][JDTP] jd9366t_GetID: IC ID: 9032
[    2.472512]  (1)[101:kworker/u16:1][JDTP] jd9366t_ReadSectionInfo: Section info was ready
[    2.476455]  (1)[101:kworker/u16:1][JDTP] FW_VER: 00009601
[    2.476469]  (1)[101:kworker/u16:1][JDTP] FW_CID_VER: 1095000f
[    2.476479]  (1)[101:kworker/u16:1][JDTP] PANEL_MAKER: INX
[    2.476490]  (1)[101:kworker/u16:1][JDTP] PANEL_VER: 01
[    2.476504]  (1)[101:kworker/u16:1][JDTP] jadard_mcu_touch_info_set: JD_X_NUM = 32, JD_Y_NUM = 48, JD_MAX_PT = 10
[    2.476515]  (1)[101:kworker/u16:1][JDTP] jadard_mcu_touch_info_set: JD_X_RES = 1200, JD_Y_RES = 2000
[    2.476526]  (1)[101:kworker/u16:1][JDTP] jadard_mcu_touch_info_set: JD_INT_EDGE = 1
[    2.476538]  (1)[101:kworker/u16:1][JDTP] jadard_chip_common_init: Use Protocol Type B
[    2.476581]  (1)[101:kworker/u16:1][JDTP] input_set_abs_params: mix_x 0, max_x 1200, min_y 0, max_y 2000
[    2.476605]  (1)[101:kworker/u16:1][JDTP] jadard_sorting_init: enter
[    2.477363]  (1)[101:kworker/u16:1][JDTP] jadard_touch_irq=18
[    2.477593]  (1)[101:kworker/u16:1][JDTP] jadard_ts_register_interrupt: irq enabled at qpio: 18
[    2.477607]  (1)[101:kworker/u16:1][JDTP] jadard_debug_init:Enter
[    2.478242]  (5)[101:kworker/u16:1][JDTP] end jadard_common_local_init, 564
```

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Appendices: Probe Failed

- **检查 I2C 节点有无建立成功**
  - 指令: ls -l sys/bus/i2c/devices
  - JD9XXXT I2C 的 Device ID 在 Device Tree 设定为0x68, 可于下图红色框处找到对应的节点

```
rk3568_r:/ # ls -l sys/bus/i2c/devices/
total 0
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 0-001c -> ../../../devices/platform/fdd40000.i2c/i2c-0/0-001c
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 0-0020 -> ../../../devices/platform/fdd40000.i2c/i2c-0/0-0020
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 1-0050 -> ../../../devices/platform/fe5a0000.i2c/i2c-1/1-0050
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 2-0068 -> ../../../devices/platform/fe5b0000.i2c/i2c-2/2-0068
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 3-0060 -> ../../../devices/platform/fe5c0000.i2c/i2c-3/3-0060
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 5-0015 -> ../../../devices/platform/fe5e0000.i2c/i2c-5/5-0015
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 5-0024 -> ../../../devices/platform/fe5e0000.i2c/i2c-5/5-0024
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 5-0024-1 -> ../../../devices/platform/fe5e0000.i2c/i2c-5/5-0024-1
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 5-0051 -> ../../../devices/platform/fe5e0000.i2c/i2c-5/5-0051
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 i2c-0 -> ../../../devices/platform/fdd40000.i2c/i2c-0
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 i2c-1 -> ../../../devices/platform/fe5a0000.i2c/i2c-1
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 i2c-2 -> ../../../devices/platform/fe5b0000.i2c/i2c-2
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 i2c-3 -> ../../../devices/platform/fe5c0000.i2c/i2c-3
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 i2c-5 -> ../../../devices/platform/fe5e0000.i2c/i2c-5
lrwxrwxrwx 1 root root 0 2022-07-14 17:36 i2c-6 -> ../../../devices/platform/fe0a0000.hdmi/i2c-6
```

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Appendices: Probe Failed

- **检查 CONFIG_JD_DB 是否有开启**
  - CONFIG_JD_DB 为JD内部高通平台开发板使用, **客戶请勿开启此设定**.
  - 若开启, TP驱动将无法Probe.

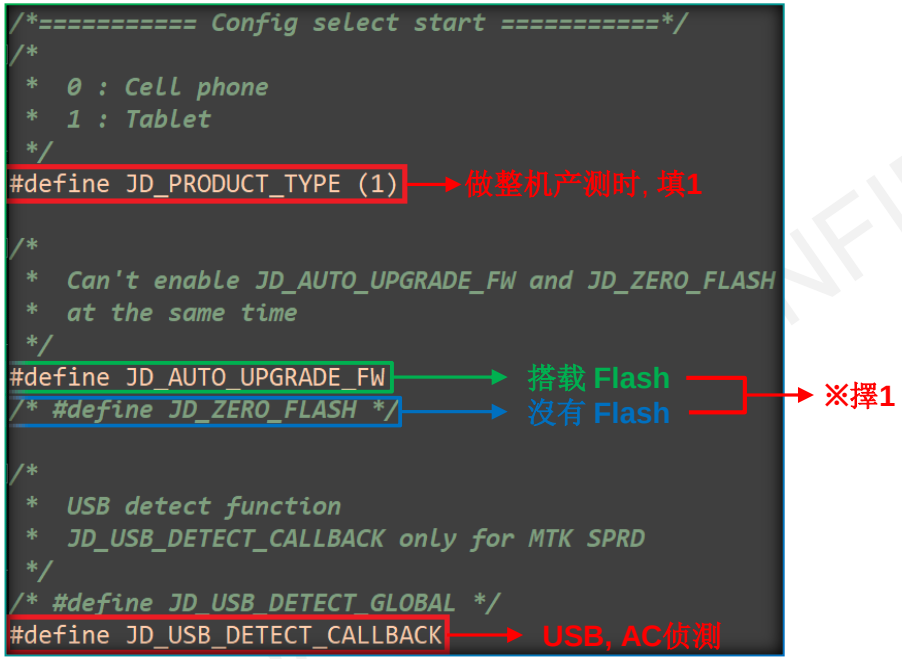
```
#if defined(CONFIG_JD_DB)
void jadard_workarround_init(void)
{
    JD_I("jadard_workarround_init by Driver\n");

    i2c_add_driver(&jadard_common_driver);
}
#endif
```

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Appendices: Function Configuration

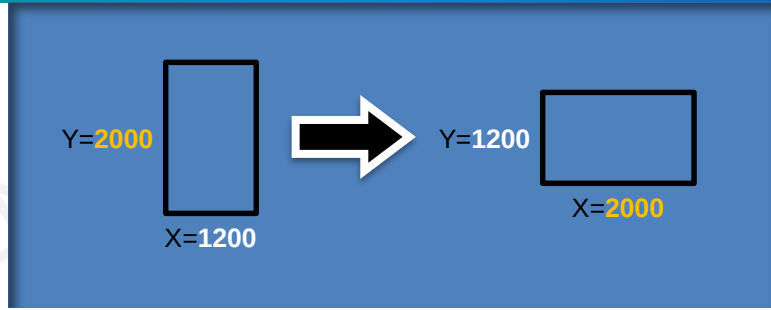
- **/kernel/drivers/input/touchscreen/jdchipset/jadard_common.h**



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Appendices: Horizontal and Vertical Direction Setting

- **由系統(平台)交換 X, Y 坐标**
  - 不需修改驱动及Touch FW



- **由驱动交换X, Y 坐标**
  - 需修改 Device Tree 的 Resolution 设定, 驱动会根据此设定交换坐标
    - ~~jadard,panel-coords = <0 1200 0 2000>;~~ 改為 jadard,panel-coords = <0 2000 0 1200>;

/kernel/drivers/input/touchscreen/jdchipset/jadard_common.c

```
    if (pjadard_ic_data->JD_X_RES <= pjadard_ic_data->JD_Y_RES) {
        input_report_abs(ts->input_dev, ABS_MT_POSITION_X, pjadard_host_data->x[i]);
        input_report_abs(ts->input_dev, ABS_MT_POSITION_Y, pjadard_host_data->y[i]);
    } else if (pjadard_ic_data->JD_X_RES > pjadard_ic_data->JD_Y_RES) {
        input_report_abs(ts->input_dev, ABS_MT_POSITION_X, pjadard_host_data->y[i]);
        input_report_abs(ts->input_dev, ABS_MT_POSITION_Y, pjadard_host_data->x[i]);
    }
```

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